

CyberShield: A 2-Tier Cascade Network Intrusion Detection System (NIDS) [cite: 1]

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Abstract—CyberShield is a 2-Tier Cascade Network Intrusion Detection System (NIDS) designed to address extreme class imbalance, dataset leakage, and real-world deployment constraints in cybersecurity systems[cite: 5, 6]. Built on the CICIDS2017 dataset, the system undergoes a rigorous data-engineering pipeline that removes duplicate records, eliminates data leakage, and fixes label inconsistencies, resulting in a clean dataset of 2,520,798 network flows[cite: 7]. The architecture consists of Tier B (Random Forest) for fast multi-class classification and Tier C (XGBoost) for deep rare-attack detection[cite: 8]. The system achieves high Macro-F1 performance while maintaining real-time inference capability[cite: 9]. Deployment is hybrid: a C++ ONNX Runtime engine runs on Android for sub-2ms edge inference, while a FastAPI service on AWS Elastic Beanstalk handles cloud requests[cite: 10].

Index Terms—Intrusion Detection System, CICIDS2017, Random Forest, XGBoost, Class Imbalance, ONNX Runtime, Edge AI, Cybersecurity [cite: 11]

I. INTRODUCTION

Modern networks generate massive traffic, requiring intelligent intrusion detection systems[cite: 12, 13]. Signature-based IDS fail against unknown attacks, motivating machine learning-based approaches[cite: 14]. However, datasets like CICIDS2017 suffer from severe class imbalance, where benign traffic dominates rare attack types such as Heartbleed and Infiltration[cite: 15]. CyberShield addresses this using a 2-Tier Cascade architecture optimized for accuracy, efficiency, and deployment feasibility[cite: 16]. Key Contributions include a cleaned CICIDS2017 dataset with leakage removal, a 2-Tier Cascade ML architecture (RF + XGBoost), and a hybrid Edge/Cloud deployment system[cite: 17, 18, 19].

II. RELATED WORK

Random Forest models are highly effective for imbalanced tabular data due to ensemble robustness[cite: 20, 21]. SMOTE introduces synthetic bias in high-dimensional datasets[cite: 22]. ONNX Runtime enables efficient cross-platform deployment[cite: 22].

III. DATASET AND DATA ENGINEERING

CICIDS2017 contains 2.83 million flow records with 78 network features[cite: 23]. Key issues identified include data leakage via the Is_Attack feature, 307,078 duplicate records, and label inconsistencies across models[cite: 24, 25]. Resolving these results in a clean dataset of 2,520,798 records[cite: 26, 30].

TABLE I
CLEAN CICIDS2017 DATASET [CITE: 27, 28, 29]

Attack Type	Count	%
BENIGN	2,018,340	80.08
DoS Hulk	205,139	8.14
PortScan	141,259	5.60
DDOS	113,705	4.51
Others	Remaining	12.00

IV. SYSTEM ARCHITECTURE

The system uses a cascading logic where Tier B predicts the initial class[cite: 42, 43]. If the confidence is below 0.70, the flow escalates to Tier C, which produces the final decision[cite: 44, 45].

Tier B (Random Forest): Configured with 75 estimators, a max depth of 25, and balanced class weighting to ensure fast general attack detection[cite: 32, 33, 34, 35, 36].

Tier C (XGBoost): Configured with 200 estimators, a learning rate of 0.1, and histogram-based training for sample-weighted rare attack detection[cite: 37, 38, 39, 40, 41].

V. DEPLOYMENT ARCHITECTURE

For edge deployment on Android, a C++ ONNX Runtime engine enables direct memory inference with sub-2ms latency and fully offline execution[cite: 46, 47, 48, 49, 50, 51]. For cloud deployment on AWS, a FastAPI microservice exposes a REST API endpoint (/classify) for scalable inference[cite: 52, 53, 54].

VI. EXPERIMENTAL RESULTS

TABLE II
MODEL PERFORMANCE COMPARISON [CITE: 55, 56, 57, 58]

Model	Acc.	Mac F1	Heartbleed	Infilt.
MLP (Leaky)	99.99*	0.0669	0.00	0.00
MLP (Clean)	82.41	0.1134	0.00	0.00
Tier B RF	99.83	0.8605	0.50	1.00
Tier C XGB	99.86	0.8821	1.00	1.00

*Accuracy inflated due to data leakage[cite: 59].

VII. CONCLUSION

CyberShield demonstrates a robust 2-Tier Cascade NIDS using Random Forest and XGBoost[cite: 60, 61]. The system achieves high accuracy, strong macro-F1 performance, and real-world deployment capability across edge and cloud environments[cite: 62].

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